



Department of Computer Science
ETH Zürich
January 6, 2025

To Whom It May Concern,

we are glad to formally acknowledge the performance of Luca Callisti, who designed and implemented the best solution for a project in Probabilistic Artificial Intelligence, a course offered in Winter 2024 at ETH Zürich.

The course focuses on core modeling techniques and algorithms from statistics, optimization, planning, and control. It touches upon topics spanning from Bayesian modeling, to active learning and reinforcement learning. Students are asked to solve several practical tasks. The student worked in a team of two students and delivered an outstanding solution for the task involving **Approximate Bayesian Inference in Neural Networks via SWA-Gaussian**. This particular project involves the implementation of Bayesian Neural Networks (BNNs) in order to classify land-use patterns from satellite images through well-calibrated, uncertainty-aware predictions.

The student proposed innovative solutions for modeling the problem, implementing and evaluating a range of stochastic weight averaging methods, while performing thorough hyperparameter tuning. Moreover, the student presented their solution clearly and concisely, delivering a great talk. For these reasons, the student's team was selected as **best among 275 teams**.

Sincerely,

Prof. Andreas Krause

ETH zürich
Probabilistic
Artificial
Intelligence **2024**

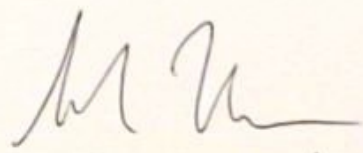
Best Project

Task 2: Approximate Bayesian
Inference in Neural Networks
via SWA-Gaussian

Awarded to:
Luca Callisti

Prof. Andreas Krause

December 18th, 2024



Learning &
Adaptive Systems